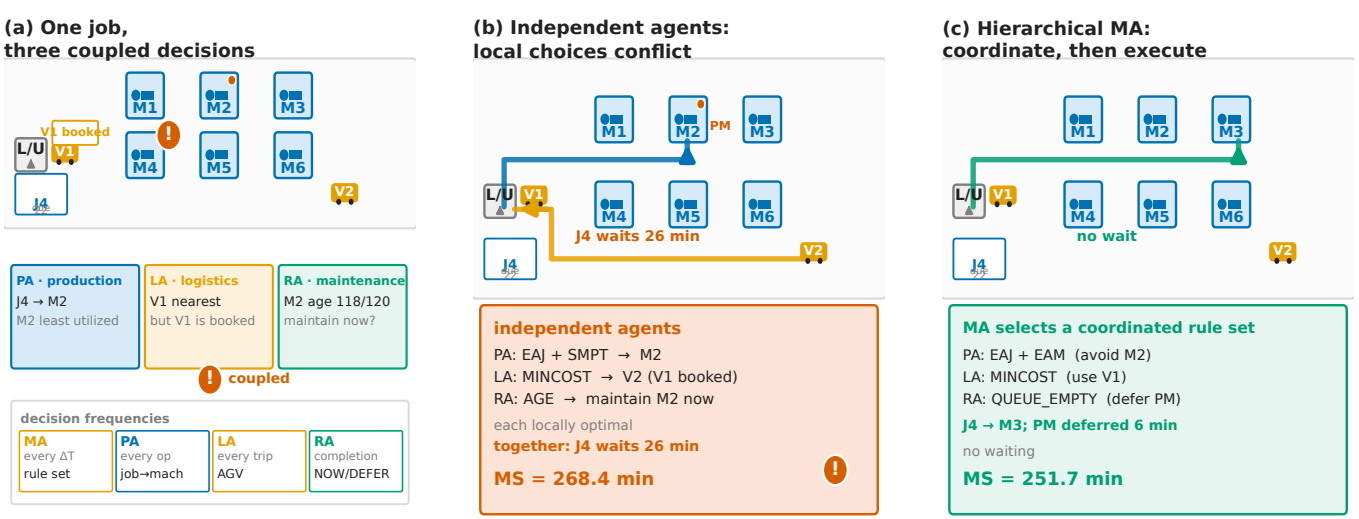

(a) One job,
three coupled decisions
V1 booked
L/U
V1
M1
M2
M3
M4
M5
M6
J4
V2
PA · production
J4 → M2
M2 least utilized
LA · logistics
V1 nearest
but V1 is booked
RA · maintenance
M2 age 118/120
maintain now?
coupled
decision frequencies
MA
every ΔT
rule set
PA
every op
job→mach
LA
every trip
AGV
RA
completion
NOW/DEFER
(b) Independent agents:
local choices conflict
M1
M2
PM
M3
L/U
V1
M4
M5
M6
J4 waits 26 min
J4
V2
independent agents
PA: EAJ + SMPT → M2
LA: MINCOST → V2 (V1 booked)
RA: AGE → maintain M2 now
each locally optimal
together: J4 waits 26 min
MS = 268.4 min
(c) Hierarchical MA:
coordinate, then execute
M1
M2
M3
L/U
V1
M4
M5
M6
no wait
J4
V2
MA selects a coordinated rule set
PA: EAJ + EAM (avoid M2)
LA: MINCOST (use V1)
RA: QUEUE_EMPTY (defer PM)
J4 → M3; PM deferred 6 min
no waiting
MS = 251.7 min